Dynamic Planet

Brown Science Olympiad Invitational
February 11, 2023



Team Name: _____

Team Number: _____

Instructions:

1. All questions are worth 1 point unless marked otherwise.
2. Put your team name and team number at the top of all answer sheets.
3. Write all answers on the answer sheets provided; answers written on the question/image sheets will not be graded.
4. Good luck!

1. Which of the following drainage patterns is not accordant?
   (a) trellis
   (b) annular
   (c) deranged
   (d) antecedent

2. Which of the following drainage patterns is commonly found around volcanoes, domes, and laccoliths?
   (a) radial
   (b) parallel
   (c) trellis
   (d) centripetal

3. Which one of the following principles would be the least helpful in determining the chronology of strata?
   (a) law of superposition
   (b) principle of cross-cutting relationships
   (c) principle of original horizonality
   (d) uniformitarian principle

4. Which of the following processes would best describe the act of ions carried in groundwater chemically precipitating to form new crystalline material between sedimentary grains?
   (a) subsidence
   (b) cementation
   (c) diapirism
   (d) cataclasis

5. Which of the following geohazards does not typically involve the act of debris flow through stream channels?
   (a) lahar
   (b) jokulhaup
   (c) mudslide
   (d) fumarole

6. Which of the following types of weathering involves physical forces that break a rock into smaller pieces?
   (a) acidic weathering
   (b) biological weathering
   (c) mechanical weathering
   (d) chemical weathering

7. Which of the following choices would most likely be used to describe the process by which overlying rock is eroded away?
   (a) sheeting
   (b) unloading
   (c) subduction
   (d) divergence

8. Which of the following choices is closest to the average percent of water in soil?
   (a) 5
   (b) 25
   (c) 50
   (d) 75

9. Which of the following choices is closest to the average percent of mineral matter in soil?
   (a) 5
   (b) 25
   (c) 50
   (d) 75

10. Which of the following choices is closest to the average percent of organic matter in soil?
    (a) 0
    (b) 5
    (c) 10
    (d) 15

11. Which of the following choices best refers to the material that covers almost all of Earth's land surface?
    (a) regolith
    (b) laccolith
    (c) baccolith
    (d) rhyolith

12. Which of the following is not a control of soil formation?
    (a) time
    (b) climate
    (c) topography
    (d) soil profile

13. Which of the following soil horizons is the topmost layer?
    (a) O horizon
    (b) E horizon
    (c) C horizon
    (d) A horizon

14. Which of the following materials is least resistant to chemical weathering?
    (a) olivine
    (b) amphibole
    (c) feldspar
    (d) quartz

15. In what climate would there most likely be the best amount of moisture to foster rapid chemical weathering?
    (a) polar
    (b) arid
    (c) continental
    (d) tropical

16. Which of the following terms is most commonly used to refer to the nonuniform weathering of rock masses?
    (a) irregular weathering
    (b) isogonal weathering
    (c) differential weathering
    (d) silicate weathering

17. What is the definition of the greek word “rhegos” that regolith is derived from?
    (a) composition
    (b) blanket
    (c) shield
    (d) dome

18. Which of the following bodies of water is comprised of freshwater?
    (a) Sea of Galilee
    (b) Sargasso Sea
    (c) South China Sea
    (d) Sea of Okhotsk

19. Which of the following is the name for the large body of water that existed where the Great Lakes are currently located?
    (a) Lake Winnipegosis
    (b) Lake Lahontan
    (c) Lake Agassiz
    (d) Lake Collier

    Lake Agassiz

20. What terms respectively describe a coast that predominantly loses sediments and a coast that predominantly gains sediments?
    (a) erosional and depositional
    (b) lossing and accumulative
    (c) negative-gain and positive-gain
    (d) contraction and expansion

21. An armored beach is protected, or "armored" with:
    (a) Moeraki boulders
    (b) lag deposits
    (c) mud balls
    (d) armored rocks

22. Bars that remain connected to the shore at one end are known as:
    (a) hanging bars
    (b) hooks
    (c) hanging spits
    (d) deltas

23. Which of the following is not a primary coast?
    (a) ria coast
    (b) fault coast
    (c) cratered coast
    (d) reef coast

24. The Western Sahara is a wind-modified depositional coast. What is the name of this type of coast?
    (a) fault coast
    (b) ria coast
    (c) dune coast
    (d) reef coast

25. Which of the following estuaries has the most uniformity over depth?
    (a) fjord-type
    (b) inverse
    (c) salt wedge
    (d) partially-mixed

26. The pink sands of beaches are usually made up of?
    (a) coral
    (b) quartz
    (c) shell material
    (d) glauconite

27. Groins act as:
    (a) buffers for seawater
    (b) structures that trap beach sediments
    (c) a restorative force for tides
    (d) sediment structures that prevent the erosion of land

28. Which of the following types of estuaries are the simplest and mostly mixed by entrainment?
    (a) salt wedge
    (b) well-mixed
    (c) fjord-type
    (d) partially-mixed

29. Flushing time can be obtained by dividing the mean volume of an estuary by
    (a) the concentration of salt
    (b) the average speed of all water
    (c) the net seaward flow
    (d) the volume of freshwater present

30. Which of the following is a structure that protects coastal areas from the force of the waves?
    (a) bulkheads
    (b) breakwaters
    (c) seawalls
    (d) groins

31. Puget Sound and the San Francisco Bay are both examples of what type of estuary?
    (a) salt wedge
    (b) well-mixed
    (c) fjord-type
    (d) partially-mixed

32. What type of coast is the Kingsbury Estuary, located in Devon, England?
    (a) fault coast
    (b) ria coast
    (c) dune coast
    (d) reef coast

33. What are the two sources of pollution and their CORRECT descriptions?
    (a) Straight-source pollution and dendritic-source pollution. Straight-source pollution can be traced to a single, discrete source while dendritic-source pollution refers to diffuse contamination.
    (b) Straight-source pollution and dendritic-source pollution. Dendritic-source pollution can be traced to a single, discrete source while straight-source pollution refers to diffuse contamination.
    (c) Point source pollution and non-point source pollution. Point source pollution can be traced to a single, discrete source while non-point source pollution refers to diffuse contamination.
    (d) Point source pollution and non-point source pollution. Non-point source pollution can be traced to a single, discrete source while point source pollution refers to diffuse contamination.

34. Name two primary chemicals that contribute to acid rain. (2 pts)

35. Describe a scenario where wet deposition of acid rain is less potent than dry deposition. (2 pts)

36. Besides climate changes (hotter temperatures, seasonal change), name one possible reason for which a river might experience thermal pollution?

37. Bob woke up from a dream. In his dream, he imagined a lake next to a farm experiencing agricultural pollution; almost no fish survive. Identify the possible sources of the pollution from the farm. Then, use the concepts of BOD and eutrophication to explain to Bob why he should convince the farmer to reduce pollution. (5 pts)

38. An aquifer experiences an evapotranspiration rate of 25.5 inches/year. It also has an infiltration rate of 37.4 inches/year, a base flow of 9.9 inches/year, and a subsea outflow of 2 inches/year. What is the aquifer's water budget? Show work and proper units. (3 pts)

39. Is the aquifer above in danger of depletion?

40. Describe how deposition might occur in a meandering channel. (2 pts)

41. Which land feature forms the Great Continental Divide?

42. What term refers to the maximum load that a stream can support?

43. What is the difference between a delta and an alluvial fan?

44. Describe where you would find the majority of coarse-grained sediment on an alluvial fan. (2 pts)

45. What is the difference between a U-shaped valley and a V-shaped valley?

46. What is another word for a watershed?

47. The phi scale is a method to classify the sizes of different types of sediments. What type of scale is it?
    (a) linear
    (b) exponential
    (c) logarithmic
    (d) nonlinear

48. Arrange the following types of sediments in order from smallest to largest under classification by the phi scale: silt, clay, sand, boulder, pebble, cobble (6 pts)

Use the topographic map found on the last page of this exam packet, to answer the following questions.

50. What sector of the topographic map is Steelhead Lake located in?

51. What type of river is Convict Creek?

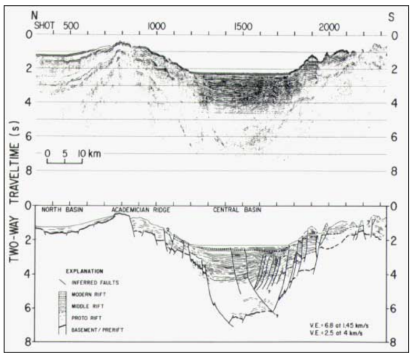
52. What cardinal direction does Hilton Creek, flow, found on the right side of the map?

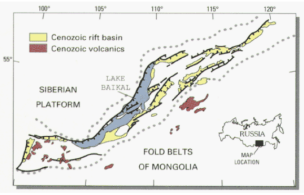
53. What geologic feature surrounds Lake Crowley, found in the top right section of the map?

54. Name 3 lakes Convict Creek flows through and in what order water passes through each of the lakes. (3 pts)

Use the diagrams below, all depicting the same lake, to answer the following questions.







55. What is the name of the lake in the images?

56. Which of the following terms best describes the formation of this lake?
    (a) Bar Built
    (b) Tectonic
    (c) Glacial
    (d) Fluvial

57. This lake is the world's largest freshwater lake by volume, containing more water than all of the North American Great Lakes combined. Approximately what percent of the world's fresh surface water is contained within this lake?
    (a) 5-10
    (b) 20-25
    (c) 35-40
    (d) 50-55

58. Name all of the Great Lakes and rank them in order of least amount of total water volume to most amount of total water volume. (5 pts)

59. This is the world's deepest lake. What is the approximate maximum depth of this lake in meters?
    (a) 1,600 m
    (b) 800 m
    (c) 200 m
    (d) 2000 m

60. This is also the world's oldest lake. Which of the following time ranges is closest to the estimated age of this lake?
    (a) 5-10 million years old
    (b) 25-30 million years old
    (c) 50-55 million years old
    (d) 95-100 million years old

ANSWER SHEET

Team Name: _____ Team Number: _____

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Team Name: _____ Team Number: _____

35. Answer below:

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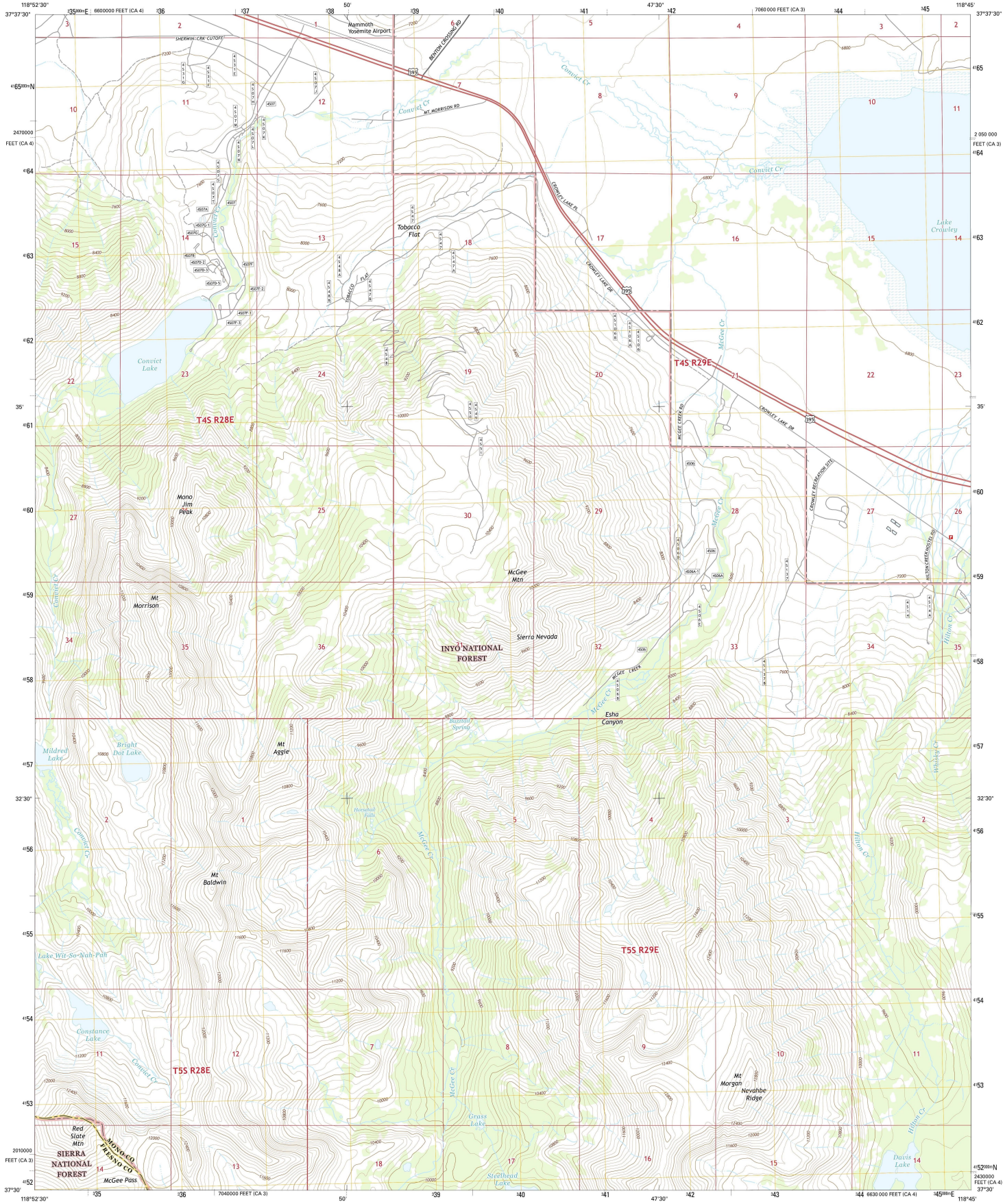
56. _____

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CONVICT LAKE QUADRANGLE
CALIFORNIA
7.5-MINUTE SERIES

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

USGS science for a changing world

The National Map US Topo

Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 11S
10 000-foot ticks: California Coordinate System of 1983 (zones 3 and 4)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, June 2012
Roads.....HERE, ©2013 - 2014
Roads within US Forest Service Lands.....FSTopo Data with limited Forest Service updates, 2012 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2012
Contours.....National Elevation Dataset, 1999
Boundaries.....Multiple sources; see metadata file 1972 - 2015
Public Land Survey System.....BLM, 2011

UTM GRID AND 2015 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
LB
Grid Zone Designation
11S

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0 0.5 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 80 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011. A metadata file associated with this product is draft version 0.6.18

CALIFORNIA
QUADRANGLE LOCATION

ADJOINING QUADRANGLES
1 Old Mammoth
2 Whitmore Hot Springs
3 Watterson Canyon
4 Bloody Mountain
5 Toms Place
6 Graveyard Peak
7 Mount Abbot
8 Mount Morgan

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route
FS Primary Route
FS Passenger Route
FS High Clearance Route

Check with local Forest Service unit for current travel conditions and restrictions.

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2015

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